		Current through cell = V/I = 9.0 /5.00 = 1.80 A
20	B	$P = \frac{E^2R}{(R+2+r)^2} = \frac{6^2R}{(R+7)^2}$ $\frac{dP}{dR} = \frac{(R+7)^2 6^2 - 2 \times 6^2 R(R+7)}{(R+7)^4}$ $= \frac{6^2(7-R)}{(R+7)^3}$ When $\frac{dP}{dR} = 0$, R = 7 Power output is maximum P_{max} = 1.29 W
21	A	The proton experiences an upward force, $F_E = qE = ma$. Hence, upward acceleration of proton, a = qE / m